

Th W13

3rd
order
ex1

$$(r-1)(r+2)(r-5) = 0$$

$$y''' - 4y'' - 7y' + 10y = 3e^{2t}$$

$$y_h = c_1 e^t + c_2 e^{-2t} + c_3 e^{5t}$$

$$y_p = a e^{2t}$$

$$y_p' = 2a e^{2t}$$

$$y_p'' = 4a e^{2t}$$

$$y_p''' = 8a e^{2t}$$

$$8a - 4(4a) - 7(2a) + 10a = 3$$

$$(8 - 16 - 14 + 10)a = 3$$

$$-12a = 3$$

$$a = \frac{3}{-12} = -\frac{1}{4}$$

$$y = c_1 e^t + c_2 e^{-2t} + c_3 e^{5t} - \frac{1}{4} e^{2t}$$

with

$$y(0) = 2, \quad y'(0) = 4, \quad y''(0) = -1$$

$$y(0) = 2$$

$$y'(0) = 4$$

$$y''(0) = -1$$

$$c_1 + c_2 + c_3 - \frac{1}{4} = 2$$

$$c_1 - 2c_2 + 5c_3 - \frac{1}{2} = 4$$

$$c_1 + 4c_2 + 25c_3 - 1 = -1$$

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$$\left[\begin{array}{ccc|c} 1 & 1 & 1 & 2 + \frac{1}{4} \\ 1 & -2 & 5 & 4 + \frac{1}{2} \\ 1 & 4 & 25 & -1 + 1 \end{array} \right]$$

3rd
order

ex 2

$$r^3 - 4r^2 - 7r + 10 = 0 \quad (r-1)(r+2)(r-5) = 0$$

$$y''' - 4y'' - 7y' + 10y = 14 \sin(3t)$$

$$y_h = c_1 e^t + c_2 e^{-2t} + c_3 e^{5t}$$

$$y_p = a \sin(3t) + b \cos(3t)$$

$$y_p' = 3a \cos(3t) - 3b \sin(3t)$$

$$y_p'' = -9a \sin(3t) - 9b \cos(3t)$$

$$y_p''' = -27a \cos(3t) + 27b \sin(3t)$$

$$y_p'''$$

$$-4y_p''$$

$$-7y_p'$$

$$+10y_p$$

$$14 \sin(3t)$$

$$-27a \cos(3t) + 27b \sin(3t)$$

$$+36b \cos(3t) + 36a \sin(3t)$$

$$-21a \cos(3t) + 21b \sin(3t)$$

$$10b \cos(3t) + 10a \sin(3t)$$

$$0 \cos(3t) + 14 \sin(3t)$$

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